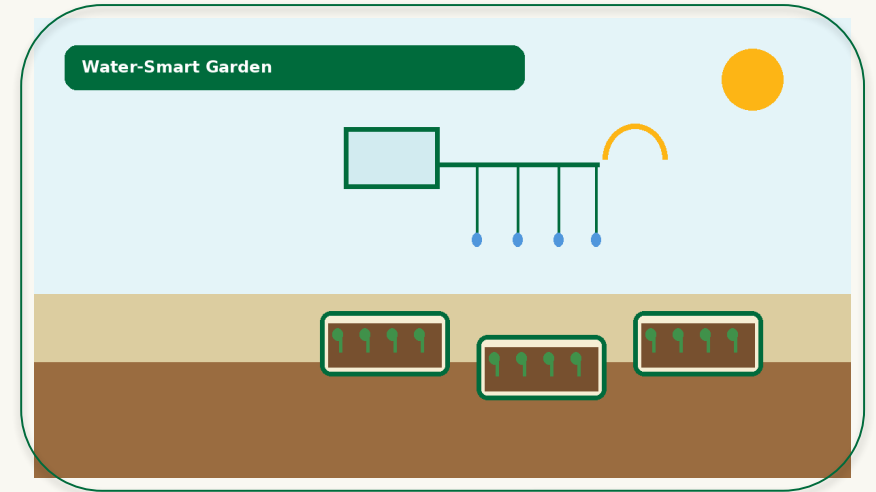


Water-Smart Garden

Using Water Efficiently

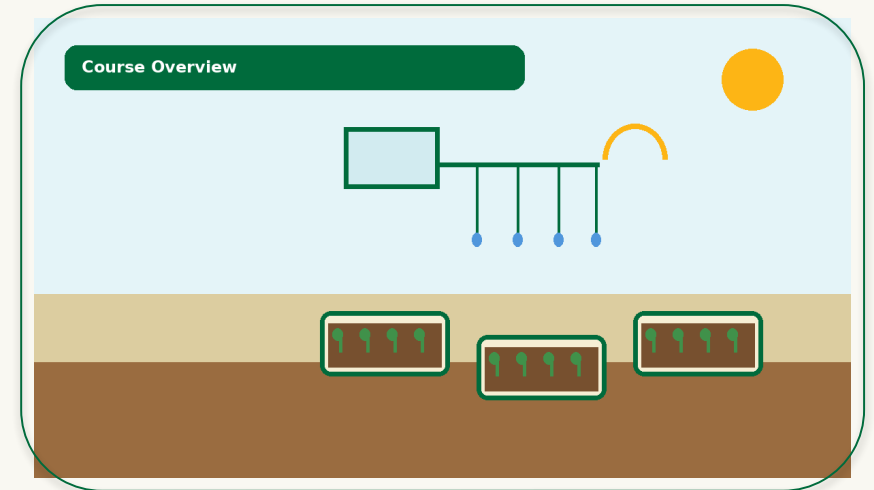
- LIFEWS AgriShine™ GrowMeal™ Primary 4-6 Teacher Training
- Audience: Primary 4-6 teachers, teaching assistants and garden facilitators
- Format: Self-paced online course + supervised garden practical
- Recommended duration: 3 hours online + 90 minutes practical
- Learning progression: measure -> experiment -> record -> analyze -> improve



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system and its educational purpose; teach all ten technical topics at Primary 4-6 level; guide safe practical work and measurement; connect science, mathematics and agriculture; maintain useful records; help pupils interpret observations without overclaiming.
- Core materials: teacher guide, system diagram, child-safe tools, labels, clipboards, rulers/tapes, graduated containers or measuring devices where relevant, observation/data sheets, clean harvest containers, handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical lesson, and submit a short lesson, measurement or investigation plan.



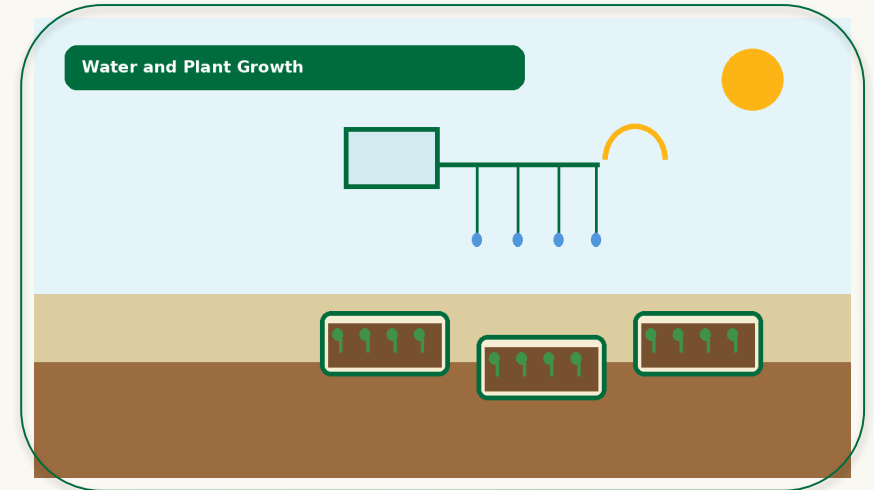
TEACHER PRACTICE

- Training sequence: EXPLAIN -> DEMONSTRATE -> MEASURE -> RECORD -> INTERPRET.

Water and Plant Growth

Teacher knowledge and practical application

- Explain that plants require water for growth and physiological processes, but needs vary with crop, stage and conditions.
- Observe signs of water stress carefully without diagnosing from one symptom alone.
- Connect watering to the root zone.
- Discuss both insufficient and excessive water.
- Use crop observations to support irrigation decisions.



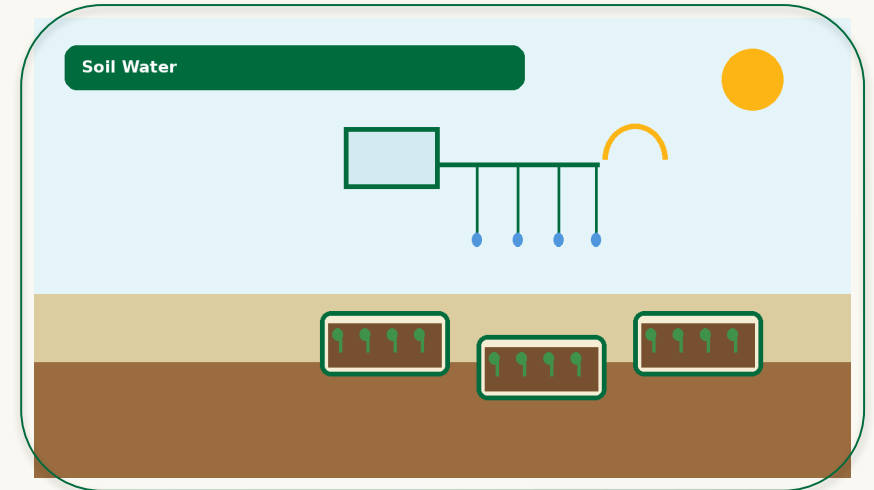
TEACHER PRACTICE

- Identify three plant observations relevant to watering decisions.

Soil Water

Teacher knowledge and practical application

- Introduce soil water as water held within pore spaces and available to roots to varying degrees.
- Compare dry, moist and very wet teacher-prepared samples.
- Use simple soil-moisture sensors if installed, alongside touch/visual observations.
- Explain that surface appearance may not represent deeper moisture.
- Avoid compacting beds during checks.



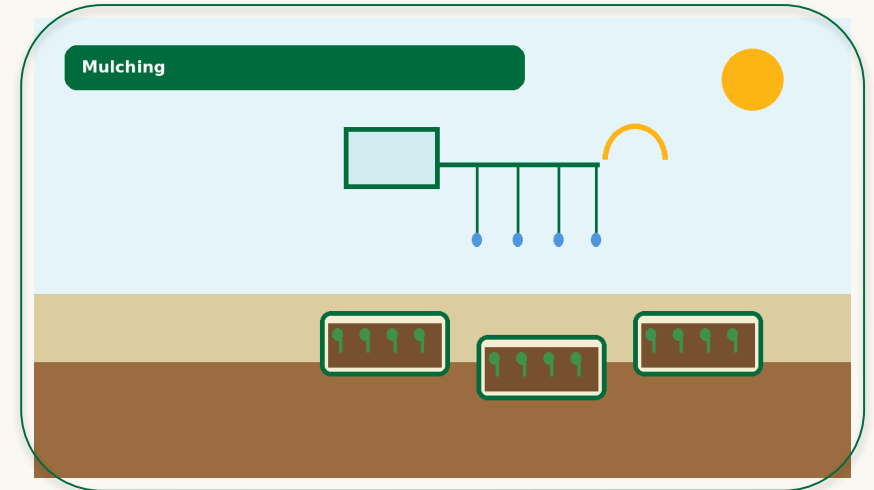
TEACHER PRACTICE

- Prepare a surface-versus-root-zone moisture check.

Mulching

Teacher knowledge and practical application

- Explain how mulch can reduce direct soil-surface evaporation and suppress some weeds.
- Use approved organic or other safe mulch materials.
- Compare mulched and unmulched areas using measured moisture or watering observations.
- Keep mulch thickness and placement consistent in comparisons.
- Monitor for pests and excessive moisture.



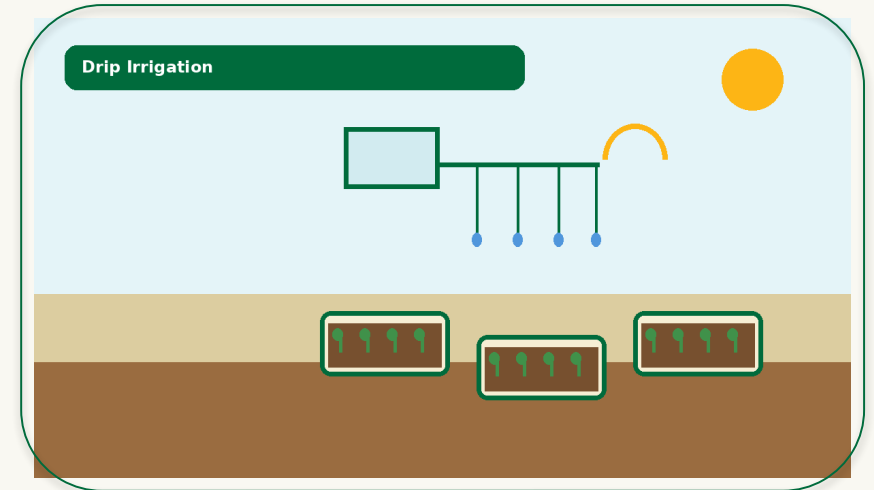
TEACHER PRACTICE

- Set up a simple mulch water-retention comparison.

Drip Irrigation

Teacher knowledge and practical application

- Identify drip components and explain targeted delivery near the root zone.
- Check emitters before watering.
- Observe wetting patterns and runoff.
- Compare drip with another approved method only under similar conditions.
- Maintain lines so efficiency is not lost through leaks or blockage.



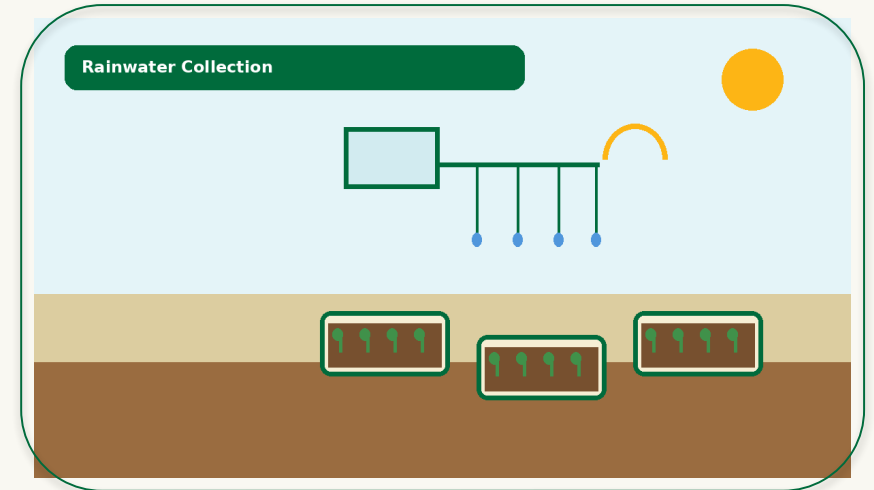
TEACHER PRACTICE

- Prepare a drip-system inspection activity.

Rainwater Collection

Teacher knowledge and practical application

- Identify roof/catchment, gutter, first-flush or screening features where installed, and storage.
- Explain that collected water quality depends on catchment and management.
- Keep storage covered and mosquito-safe.
- Do not use untreated harvested rainwater for drinking.
- Follow school/local requirements for its permitted garden use.



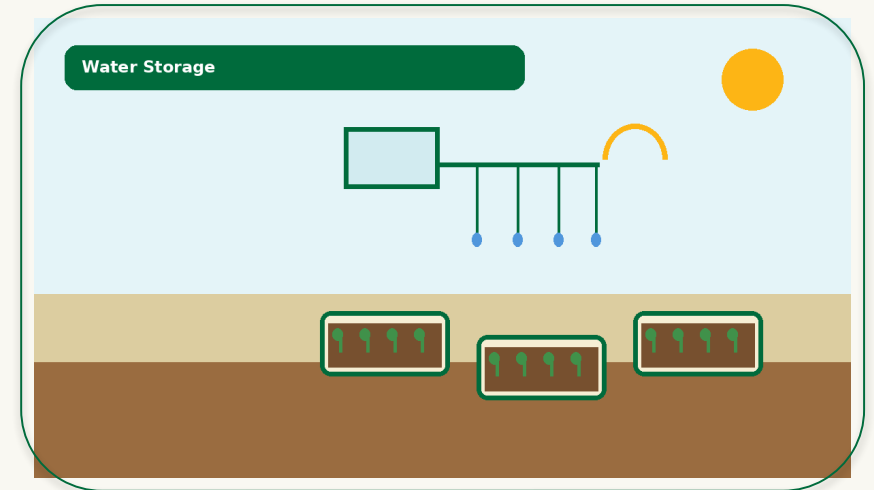
TEACHER PRACTICE

- Trace the rainwater path from catchment to storage.

Water Storage

Teacher knowledge and practical application

- Discuss why stored water can help separate collection time from irrigation time.
- Identify tank capacity, level indicator, outlet and overflow.
- Keep tanks covered, structurally secure and inaccessible for climbing.
- Monitor leaks and overflow.
- Record approximate stored volume where a calibrated indicator exists.



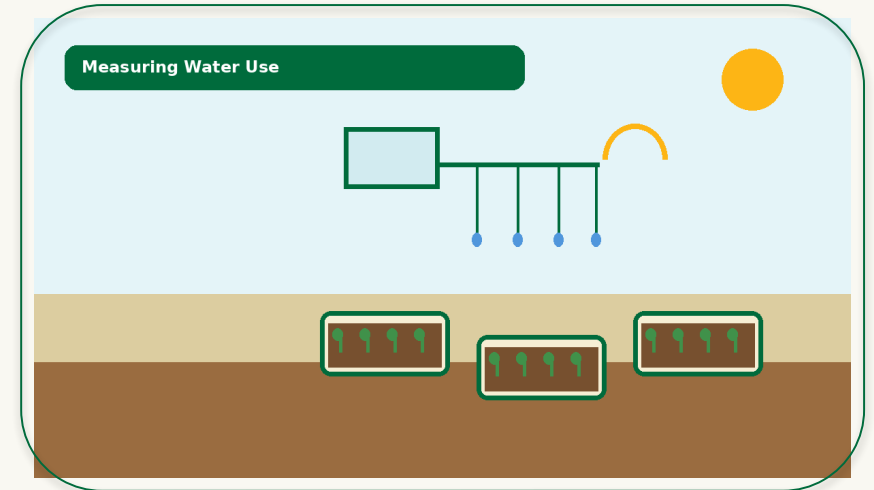
TEACHER PRACTICE

- Prepare a tank-level recording exercise.

Measuring Water Use

Teacher knowledge and practical application

- Measure irrigation using graduated containers, flow meters or known watering-can volumes.
- Record volume, crop/bed and date.
- Use consistent units.
- Relate water applied to bed area only when measurements support the calculation.
- Check records before comparing treatments.



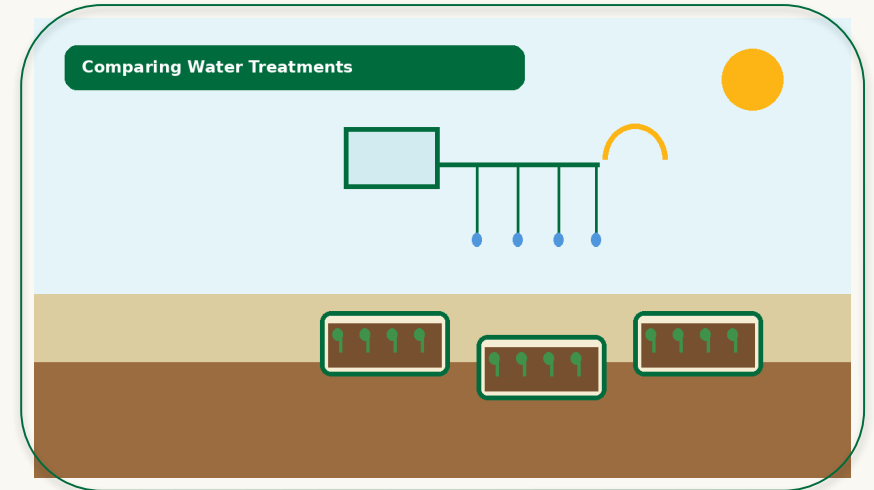
TEACHER PRACTICE

- Create a weekly water-use table.

Comparing Water Treatments

Teacher knowledge and practical application

- Set up safe treatments such as mulch versus no mulch or two approved irrigation schedules.
- Keep crop, plot size and measurement methods as similar as practical.
- Measure both water applied and a relevant plant/soil response.
- Use replication where space allows.
- Avoid deliberately stressing food crops beyond safe educational limits.



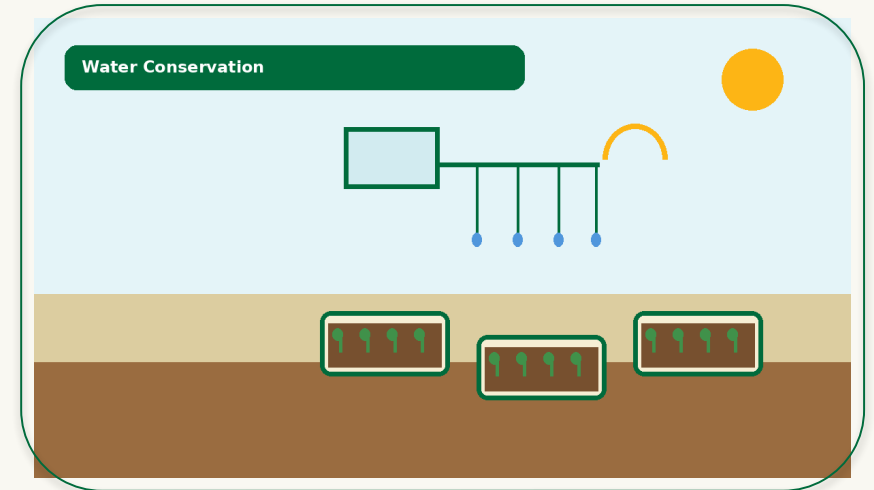
TEACHER PRACTICE

- Design a two-treatment water experiment.

Water Conservation

Teacher knowledge and practical application

- Teach conservation as reducing unnecessary losses while meeting crop needs.
- Identify leaks, overwatering, runoff, poor timing and blocked emitters as manageable losses.
- Use mulch, targeted irrigation and maintenance appropriately.
- Do not equate conservation with simply applying the least water possible.
- Connect saved water to reliable garden operation.



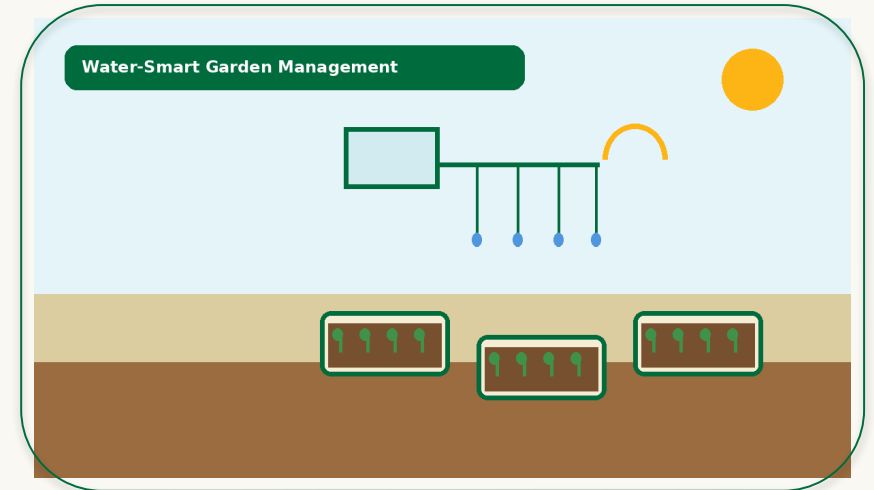
TEACHER PRACTICE

- Conduct a water-loss walk-through.

Water-Smart Garden Management

Teacher knowledge and practical application

- Combine moisture checks, irrigation records, maintenance and crop observations in one routine.
- Define who checks the system and when.
- Use data to adjust watering within teacher-approved limits.
- Review water use alongside crop condition and harvest.
- Maintain safe, dry pathways and secure storage.



TEACHER PRACTICE

- Completion task: create a weekly water-smart management plan.